Supplementary material: A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video

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Appendices A–C accompany the main manuscript. Equation numbering continues from the main manuscript.

Appendix A: Complete Specification of LOMM

This appendix gives the formal LOMM specification behind the conceptual account in the main text. It is intended for readers who want to reimplement the generator or examine its assumptions and identification limits.

A.1 The Latent Transition

The transition below advances the latent level and velocity over one observation interval while holding the acceleration input constant. The matrix $\mathbf{\Phi}_f(\Delta)$ describes how the current state evolves, and $\mathbf{\Gamma}_f(\Delta)$ describes how the input changes that state. For the oscillator in Equation 1 of the main manuscript, these are

$$\mathbf{\Phi}_f(\Delta) = \exp(\mathbf{A}_f\Delta), \qquad \mathbf{A}_f = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix} \qquad \text{and} \qquad \mathbf{\Gamma}_f(\Delta) = \int_0^\Delta \exp(\mathbf{A}_f u)\mathbf{b}\, du, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the nonzero position-feedback values used in the simulation grids, the input map has the equivalent exact form

$$\mathbf{\Gamma}_f(\Delta) = \mathbf{A}_f^{-1}\{\mathbf{\Phi}_f(\Delta) - \mathbf{I}_2\}\mathbf{b}.$$

The implemented state recurrence is

$$\begin{bmatrix} x_{if,n+1} \\ v_{if,n+1} \end{bmatrix} = \mathbf{\Phi}_f(\Delta) \begin{bmatrix} x_{if,n} \\ v_{if,n} \end{bmatrix} + \mathbf{\Gamma}_f(\Delta) q_{if,n}, \qquad n = 1, \ldots, T-1.$$

The integral definition remains valid when $\mathbf{A}_f$ is singular, including $\eta_f = 0$. If a root-based closed form is used numerically, a separate repeated-root branch is required when $\zeta_f^2 + 4\eta_f$ is close to zero.

The state components $x_{if,n}$ and $v_{if,n}$ are the level and its rate of change at t_n; η_f is position feedback, ζ_f is velocity feedback, and σ_{qf} is the standard deviation of the acceleration input. On the interval $[t_n, t_{n+1})$ the underlying continuous-time model is $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if,n}$ with $q_{if,n}$ held constant, so $\mathbf{\Phi}_f(\Delta)$ propagates the deterministic state and $\mathbf{\Gamma}_f(\Delta)$ loads the input.

Let $\mathbf{q}_{i,n} = (q_{i1,n}, \ldots, q_{im,n})^\top$ and $\mathbf{D}_q = \mathrm{diag}(\sigma_{q1}, \ldots, \sigma_{qm})$. The generator draws

$$\mathbf{q}_{i,n} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D}_q \mathbf{R}_q \mathbf{D}_q)\,.$$

The primary designs use $\mathbf{R}_q = \mathbf{I}_m$. The correlated-input sensitivity uses $\mathbf{R}_q = (1 - 0.15)\mathbf{I}_m + 0.15\mathbf{1}_m\mathbf{1}_m^\top$. Table B2 reports the Study 2 values of σ_{qf}.

In Study 1, σ_{qf} was set so that the stationary variance of $x_{if,n}$ was one. Specifically, if $\mathbf{P}_f^{(1)}$ solves $\mathbf{P}_f^{(1)} = \mathbf{\Phi}_f \mathbf{P}_f^{(1)} \mathbf{\Phi}_f^\top + \mathbf{\Gamma}_f \mathbf{\Gamma}_f^\top$, then $\sigma_{qf} = \{\mathbf{e}_1^\top \mathbf{P}_f^{(1)} \mathbf{e}_1\}^{-1/2}$.

Inputs are independent across clips and transitions and are independent of initial states, activity schedules, and measurement errors. They are also independent across dimensions when $\mathbf{R}_q = \mathbf{I}_m$. Each acceleration input is held constant for a full observation interval, a convention called zero-order hold. The input is therefore defined per interval; it is not a continuous-time white-noise process. The covariance added in one transition is $\mathbf{\Gamma}_f(\Delta)\sigma_{qf}^2\mathbf{\Gamma}_f(\Delta)^\top$ for a single dimension.

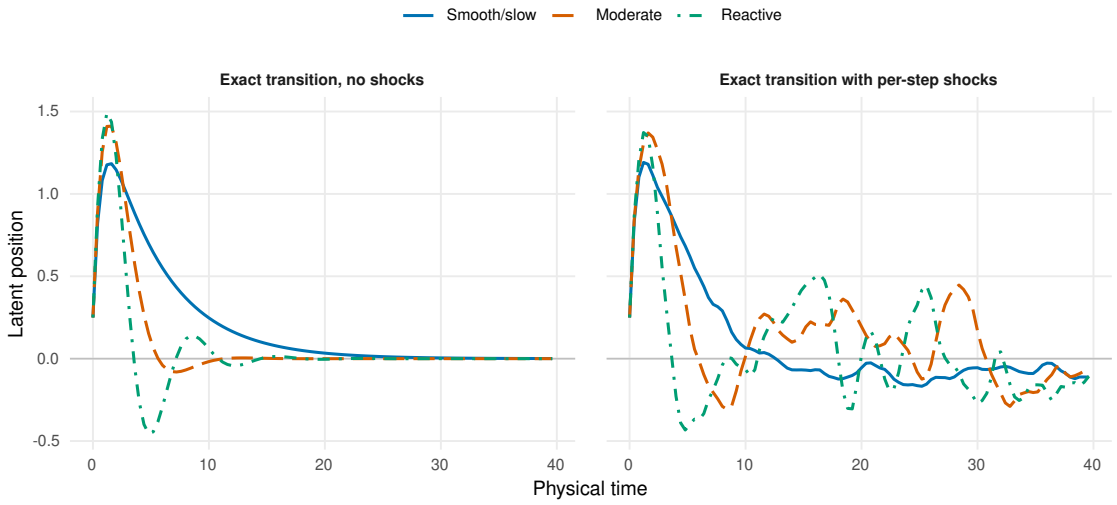


Figure A1: Illustrative latent trajectories for three uncoupled dimensions under the exact transition, shown without and with piecewise-constant acceleration inputs. The inputs create irregular peaks and reversals while the feedback parameters govern the underlying dynamic regime.

A.2 Dynamic Regimes and Finite-Horizon Amplification

The characteristic roots describe whether the deterministic motion decays, grows, or oscillates. For dimension f, they are

$$r_{f,\pm} = \frac{\zeta_f \pm \sqrt{\zeta_f^2 + 4\eta_f}}{2}.$$

Asymptotic stability means that the effect of an initial state eventually decays to zero in the absence of new inputs. In continuous time, it requires $\eta_f < 0$ and $\zeta_f < 0$, and the roots are oscillatory when $\zeta_f^2 + 4\eta_f < 0$. For the sampled transition, the spectral radius ρ is the largest absolute eigenvalue of $\mathbf{\Phi}_f(\Delta)$. Stability requires

$$\rho\{\mathbf{\Phi}_f(\Delta)\} = \exp[\Delta \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}] < 1.$$

The real parts of the characteristic roots determine whether the effect of an initial state decays or grows. Over the observation horizon $H = (T-1)\Delta$, we summarize this modal change by

$$g_f(H) = \exp[H \max\{\mathrm{Re}(r_{f,+}), \mathrm{Re}(r_{f,-})\}]\,.$$

This quantity describes deterministic modal growth or decay. It does not bound individual stochastic trajectories or capture all possible transient amplification. Table C6 reports the maximum real part of the roots and the modal factors, while Table C5 summarizes the amplitudes actually produced by the generator. Activity modulation can create additional nonstationarity and is reported separately. The amplifying configurations are finite-record stress tests.

A.3 Initial States

The system is uncoupled: η_f, ζ_f, and σ_{qf} may vary by dimension, and a dimension does not directly change the level or velocity of another dimension.

Study 1 initialized every dimension independently at the start of each uninterrupted segment by drawing $\mathbf{s}_{if,1} = (x_{if,1}, v_{if,1})^\top$ from its zero-mean stationary Gaussian distribution. A new state was drawn after each interruption.

Study 2's primary simulation design initialized every dimension in every clip at $x_{if,1} = 0.25$ and $v_{if,1} = 2$. The sensitivity analysis replaced these common fixed values with draws that were independent across dimensions and clips.

For a stable dimension, the stationary covariance $\mathbf{P}_f$ solves

$$\mathbf{P}_f = \mathbf{\Phi}_f(\Delta)\mathbf{P}_f\mathbf{\Phi}_f(\Delta)^\top + \mathbf{\Gamma}_f(\Delta)\sigma_{qf}^2\mathbf{\Gamma}_f(\Delta)^\top,$$

whose solution is unique when $\rho\{\mathbf{\Phi}_f(\Delta)\} < 1$. Drawing $\mathbf{s}_{if,1} \sim \mathcal{N}(\mathbf{0}, \mathbf{P}_f)$ makes the state process stationary

before the activity layer is applied; time-varying activity modulation still renders $a_{if}(t)x_{if}(t)$ nonstationary.

For a dimension with positive velocity feedback, the sensitivity draws $x_{if,1} \sim \text{Unif}(-0.25, 0.25)$ and $v_{if,1} \sim \text{Unif}(-2, 2)$. The alternative therefore changes cross-dimension and cross-clip coherence, marginal means, covariance, stationarity, and the magnitude and direction of the initial transient; the observed horizon is unchanged.

A.4 The Activity Layer at a Transition

Changes in an observed indicator can arise from changes in the latent level or changes in its activity multiplier. For the expressed signal $\tilde{x}_{if,n} = a_{if,n}x_{if,n}$,

$$\frac{\tilde{x}_{if,n+1} - \tilde{x}_{if,n}}{\Delta} = a_{if,n+1}\frac{x_{if,n+1} - x_{if,n}}{\Delta} + \frac{a_{if,n+1} - a_{if,n}}{\Delta}x_{if,n}.$$

The first term is latent change expressed at the new activity level. The second is change caused by the activity multiplier. An abrupt activity transition can therefore contribute strongly to an estimated derivative even though the latent level remains continuous. The gradual-transition sensitivity analysis examines whether recovery depends on these abrupt changes. At each observation, activity multiplies the completed latent state at the same observation index.

The activity schedule takes two nonnegative values, a_{high} and a_{low}. In non-switching families the dominant dimension remains at a_{high} for the whole record and the others remain at a_{low}. In switching families, dwell fractions determine how long each dimension remains dominant before the schedule reassigns a_{high}. Appendix B lists the numeric gains and dwell values used in the Study 2 families.

A.5 The Observation Layer

LOMM has two observation models. The continuous-indicator model maps latent processes directly to observed variables with specified loadings and measurement error. The bounded-score model first generates log-ratio coordinates and then converts them to positive scores that sum to one. Its ILR coordinates combine the logarithms of multiple score channels; there is no separate ILR coordinate for each channel. We use separate notation for continuous indicators, $\mathbf{z}$, and log-ratio coordinates, $\mathbf{u}$.

Let J denote the number of positive score channels, let $d = J - 1$, and let $\mathbf{V} \in \mathbb{R}^{J \times d}$ be a fixed orthonormal isometric log-ratio (ILR) basis satisfying $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_d$ and $\mathbf{V}^\top\mathbf{1}_J = \mathbf{0}$. The Study 1 basis used here is a normalized Helmert basis constructed in the stated 11-channel order. The full observation model in log-ratio

coordinates is

$$\mathbf{u}_{in} = \boldsymbol{\mu}^{(u)} + \mathbf{b}_i^{(u)} + \kappa s_i^{(u)} \left(\boldsymbol{\Lambda}_i^{(u)} \tilde{\mathbf{x}}_{in} + \boldsymbol{\varepsilon}_{in}^{(u)} \right).$$

Here $\mathbf{u}_{in}, \boldsymbol{\mu}^{(u)}, \mathbf{b}_i^{(u)}, \boldsymbol{\varepsilon}_{in}^{(u)} \in \mathbb{R}^d$, $s_i^{(u)} > 0$ is the clip scale, $\kappa > 0$ is a global scale, and $\boldsymbol{\Lambda}_i^{(u)} \in \mathbb{R}^{d \times m}$ maps the m activity-scaled latent positions into the log-ratio coordinates. The residual covariance is $\boldsymbol{\Sigma}_i^{(u)} \in \mathbb{R}^{d \times d}$, with $\boldsymbol{\varepsilon}_{in}^{(u)} \mid \boldsymbol{\Sigma}_i^{(u)} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_i^{(u)})$. The clip scale multiplies both signal and residual, and the baseline is additive.

The continuous-indicator branch has output dimension p and uses separate measurement parameters:

$$\mathbf{z}_{in} = \mathbf{b}_i^{(z)} + s_i^{(z)} \left(\boldsymbol{\Lambda}_i^{(z)} \tilde{\mathbf{x}}_{in} + \boldsymbol{\varepsilon}_{in}^{(z)} \right), \qquad \boldsymbol{\varepsilon}_{in}^{(z)} \mid \boldsymbol{\Psi}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_i),$$

where $\mathbf{z}_{in}, \mathbf{b}_i^{(z)}, \boldsymbol{\varepsilon}_{in}^{(z)} \in \mathbb{R}^p$, $\boldsymbol{\Lambda}_i^{(z)} \in \mathbb{R}^{p \times m}$, $\boldsymbol{\Psi}_i \in \mathbb{R}^{p \times p}$, and $s_i^{(z)} > 0$. In both branches, residuals are conditionally independent over time and across clips and are independent of latent states, initial states, activity schedules, and acceleration inputs. Designs with fixed measurement parameters use a common loading matrix and residual covariance within the relevant branch.

The residual-correlation sensitivity condition applies to the continuous indicators and sets

$$\boldsymbol{\Psi} = s_\varepsilon^2 \left\{ (1 - \rho) \mathbf{I}_p + \rho \mathbf{1}_p \mathbf{1}_p^\top \right\}, \qquad \rho = 0.10,$$

which introduces residual covariance between every pair of the p indicators, including pairs from different prespecified blocks.

The baseline constructions are also branch-specific. Study 1 draws $\mathbf{b}_i^{(u)}$ from a zero-mean Gaussian whose covariance is estimated from MAFW calibration videos in the full d-dimensional log-ratio coordinate space. Study 2's between-clip-variability analysis uses

$$\mathbf{b}_i^{(z)} = \boldsymbol{\Lambda}^{(z)} \boldsymbol{\beta}_i, \qquad E(\boldsymbol{\beta}_i) = \mathbf{0},$$

so that between-clip variation stays within the column space of the continuous-indicator loading matrix. Study 2 conditions without baseline and scale differences set $\mathbf{b}_i^{(z)} = \mathbf{0}$ and $s_i^{(z)} = 1$, giving

$$\mathbf{z}_{in} = \boldsymbol{\Lambda}_i^{(z)} \tilde{\mathbf{x}}_{in} + \boldsymbol{\varepsilon}_{in}^{(z)}.$$

This equation defines the continuous indicators used in Study 2's primary recovery designs before any bounded-score mapping. Each of the $p = mk$ indicators loads on one dimension and has diagonal measurement error. In the matched-method and extended factorial designs, each primary loading receives an independent Unif$(-0.15, 0.15)$ deviation from its nominal value, and each residual variance receives an independent Unif$(-0.01, 0.01)$ deviation from s_ε^2. The one-at-a-time sensitivity and between-clip-variability designs use fixed equal values except where the stated condition changes them. Zero baseline and unit scale remove baseline and overall-scale differences; common loadings and residual covariance require the separate fixed-parameter condition.

For bounded output, LOMM converts the coordinates to strictly positive scores that sum to one, the open probability simplex. The centered log-ratio (CLR) expresses each channel relative to the geometric mean of all channels. The isometric log-ratio (ILR) represents the same information in $d = J - 1$ orthonormal coordinates. Formally,

$$\mathrm{clr}(\mathbf{p}) = \log \mathbf{p} - J^{-1}\{\mathbf{1}_J^\top \log(\mathbf{p})\}\mathbf{1}_J, \qquad \mathrm{ilr}_{\mathbf{V}}(\mathbf{p}) = \mathbf{V}^\top \mathrm{clr}(\mathbf{p}),$$

with the logarithm applied componentwise. For the canonical bounded LOMM output, applying softmax to $\mathbf{V}\mathbf{u}_{in}/\tau$ gives

$$\mathrm{ilr}_{\mathbf{V}}(\mathbf{p}_{in}) = \frac{\mathbf{u}_{in}}{\tau}, \qquad \mathrm{clr}(\mathbf{p}_{in}) = \frac{\mathbf{V}\mathbf{u}_{in}}{\tau},$$

The second identity follows because the components of $\mathbf{V}\mathbf{u}_{in}$ sum to zero. The fixed basis excludes the unidentified common-logit direction and the softmax produces strictly positive rows summing to one, so the scores identify the scaled coordinates $\mathbf{u}_{in}/\tau$ exactly. A common-logit component remains unobservable, and the unscaled coordinates and τ are not separately identified without an external scale constraint.

The Study 2 bounded-score sensitivity condition uses the $p = mk$ continuous-indicator vector directly. With $J = p$, it computes

$$\mathbf{p}_{in}^{\mathrm{dep}} = \mathrm{softmax}\left(\frac{\mathbf{z}_{in}}{\tau}\right), \qquad \mathrm{clr}\left(\mathbf{p}_{in}^{\mathrm{dep}}\right) = \frac{1}{\tau}\mathbf{P}_p\mathbf{z}_{in}, \qquad \mathbf{P}_p = \mathbf{I}_p - \frac{\mathbf{1}_p\mathbf{1}_p^\top}{p}.$$

For any orthonormal ILR basis $\mathbf{V}_p \in \mathbb{R}^{p\times(p-1)}$, $\mathbf{V}_p\mathbf{V}_p^\top = \mathbf{P}_p$. Defining $\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{V}_p^\top\mathbf{z}_{in}$ therefore gives $\mathbf{V}_p\mathbf{u}_{in}^{\mathrm{dep}} = \mathbf{P}_p\mathbf{z}_{in}$. Softmax is invariant to the removed row mean, so the direct-softmax branch is equivalent to applying the basis form to $\mathbf{u}_{in}^{\mathrm{dep}}$. The analysis used the p raw scores and the p CLR components, which

satisfy a zero-sum constraint; it did not analyze the $(p-1)$ ILR coordinates. The common-logit direction in $\mathbf{z}_{in}$ is unidentifiable after this transformation.

In the equal-loading reference condition used for this sensitivity, an equal increase in all expressed latent levels adds the same amount to every logit. Softmax removes this common component. Specifically, $\mathbf{\Lambda}^{(z)}\mathbf{1}_m = \lambda\mathbf{1}_p$, so $\mathbf{P}_p\mathbf{\Lambda}^{(z)}\mathbf{1}_m = \mathbf{0}$. The projected loading matrix $\mathbf{P}_p\mathbf{\Lambda}^{(z)}$ has rank $m-1$, leaving one fewer independent latent direction. The transformation therefore changes the latent representation as well as the associations among indicators. The original channel grouping may still contain recoverable information; this sensitivity analysis tests whether methods recover that grouping after the transformation.

A.6 Converting Period and Damping to Feedback Parameters

The Study 1 calibration search specified dynamics by natural period and damping ratio. The following equations convert those settings to the feedback parameters η_f and ζ_f. If P_f is the natural period in observation intervals and ξ_f is the damping ratio, then

$$\omega_{n,f} = \frac{2\pi}{P_f\Delta}, \qquad \eta_f = -\omega_{n,f}^2, \qquad \zeta_f = -2\xi_f\omega_{n,f},$$

and the damped angular frequency is $\omega_{d,f} = \omega_{n,f}\sqrt{1-\xi_f^2}$. With the Study 1 settings $P_f = 12$, $\xi_f = 0.20$, and $\Delta = 1$, these give $\eta_f = -0.2742$ and $\zeta_f = -0.2094$. The acceleration inputs were scaled to give unit stationary position variance.

A.7 Identifiability

The simulations fix the generating parameters in advance. Estimating them from observed data would require additional constraints because different parameter combinations can produce the same observations. The main ambiguities concern scale, dimension labels, and sampling frequency.

For either observation branch $h \in \{u, z\}$ and any nonsingular diagonal matrix $\mathbf{D}$,

$$\mathbf{\Lambda}^{(h)}\tilde{\mathbf{x}}_{in} = (\mathbf{\Lambda}^{(h)}\mathbf{D}^{-1})(\mathbf{D}\tilde{\mathbf{x}}_{in}),$$

so loading magnitude and latent-trajectory scale are not jointly identified without an explicit scale constraint. Activity gain and acceleration-input scale contribute to latent amplitude in the same way. The softmax layer adds invariance to a common location shift and a tradeoff between the temperature and the coordinate scale.

Dimension labels are recoverable only up to permutation.

Sampling creates another ambiguity: different oscillation frequencies can produce the same values at the observation times. This is called aliasing. For an underdamped dimension, write

$$r_{f,\pm} = \frac{\zeta_f}{2} \pm i\omega_f, \qquad \omega_f = \frac{1}{2}\sqrt{-(\zeta_f^2 + 4\eta_f)}.$$

At observation times $t_n = (n-1)\Delta$, the frequencies ω_f and $\omega_f + 2\pi q/\Delta$ for integer q generate the same sampled sine and cosine sequences after an appropriate change in initial velocity. Identifying continuous-time oscillator parameters from data therefore requires a cadence-dependent restriction such as

$$0 < \omega_f < \frac{\pi}{\Delta},$$

together with a specified acceleration-input model and constraints on the latent state basis. Together, these ambiguities require sign and scale constraints on the dimensions, normalization of the activity schedule, anchoring of either the loading scale or the input scale, a reference-logit or sum-to-zero constraint, a fixed or otherwise identified temperature, and this frequency restriction.

Every oscillatory mode used here satisfies the frequency restriction. In Study 1, the selected damped angular frequency is approximately 0.513 radians per observation; the largest frequency in the calibration search is approximately 0.524. Both are below the sampling bound of π. Study 2 uses $\Delta \in \{0.25, 0.4\}$, giving bounds of approximately 12.57 and 7.85 against a largest angular frequency of approximately 1.04. The simulations do not estimate any of these parameters.

Appendix B: Study Designs and Implementation Details

This appendix gives the design and implementation details for both studies. Study 1 evaluates resemblance to real bounded-score series. Study 2 evaluates recovery of a prespecified factor structure from continuous indicators.

B.1 Study 1 Scoring and Calibration

The sample contains 100 MAFW videos selected across the available expression categories. Category labels served only to stratify sampling and fold construction. The analysis retained 3,557 complete observations in 188 uninterrupted segments. Individual videos contain 16 to 50 complete observations, with a median of 35. Segment lengths range from 1 to 50 observations, with a median of 13.5.

transforEmotion sampled up to 50 frames uniformly from each video and used OpenAI CLIP ViT-Large/14. A Haar cascade detected faces, and the largest detected face was expanded by 50 pixels on each side before classification. Frames without a detected face received missing scores. Eleven lowercase labels were passed directly to the text encoder: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. Burned-in subtitles were not masked when they fell inside the padded face crop.

Scores were required to lie in $[0, 1]$. A complete row was retained when its sum was positive, finite, and within 10^{-6} of one, then divided by that sum in the fixed channel order. Values were not truncated, and zero replacement was disabled. The transformation stops with an error if a complete row contains a zero. None of the 39,127 retained scores equaled zero; the minimum before row normalization was approximately 2.42×10^{-5}. The analysis used complete observations and preserved the uninterrupted segment boundaries.

Features and segment boundaries. The evaluation panel contains eleven statistics: median level, interquartile range of levels, interquartile range of changes, lag-1 autocorrelation, run-length entropy, sample entropy, permutation entropy, spectral entropy, effective rank, robust level variance, and robust change variance. The eight univariate statistics were computed for each of the eleven score channels and each of their eleven CLR coordinates. The three joint statistics used the score panel and its ten ILR coordinates. For each video, the median of the finite channel values summarized each univariate statistic. Together with the joint statistics, these summaries give 22 features. A feature remained undefined if no channel had a finite value.

Levels were pooled across all retained observations within a video. Changes and lagged pairs were formed within segments and then pooled across the video; no pair crossed a segment boundary. Lag-1 correlation was computed from these pooled pairs. Run lengths were formed separately within segments, after removing

zero changes, capped at eight, and pooled across segments. Sample and permutation entropy pooled windows formed entirely within segments. Sample entropy compared these valid windows across the video, including windows from different segments. Spectral entropy was computed separately for each eligible segment and averaged using segment length minus one as the weight. A segment containing one observation contributed to level statistics but supplied no changes, lagged pairs, or entropy windows.

Change IQR and robust change variance required at least one within-segment change. Lag-1 correlation required at least two pairs and nonzero variation in the first and second observations in the pairs. Run-length entropy required four observations across the video and was zero when fewer than two runs remained.

Sample entropy used order 2, windows of three observations, and a tolerance of 0.20 times the channel's standard deviation across the video. It required at least two windows and used the negative logarithm of $(A+0.5)/(B+1)$, where A counts matching pairs of three-observation windows and B counts matches between their first two observations; a zero or undefined channel standard deviation produced zero. Permutation entropy used order 3, required at least two windows, and resolved ties by position within the window.

Spectral entropy required segments of at least eight observations with positive total spectral power and at least two positive frequency powers. Statistics failing their requirements remained undefined.

Effective rank used the positive eigenvalues of the video correlation matrix and required that matrix to be finite. Robust level variance was the 10% trimmed mean squared distance from the video center divided by the coordinate dimension. Robust change variance was the 10% trimmed mean squared within-segment first difference divided by twice the coordinate dimension. The joint statistics required at least two observations.

Calibration. The 100 videos were divided into four category-stratified folds. Each fold used 75 calibration videos and 25 evaluation videos. The calibration videos determined the coordinate center, loading subspace, between-video baseline covariance, and empirical distribution of within-video scale. The first three eigenvectors of the pooled within-video covariance supplied the loading directions. The between-video baseline covariance was shrunk halfway toward its diagonal, and its eigenvalues were floored at 10^{-8}. The generator included between-video baseline and scale variation. Activity modulation was disabled: every latent dimension had an activity multiplier of one at every generated observation.

Generator settings were selected within each calibration set. The dynamic search crossed periods of 12, 24, and 48 observations with damping ratios of 0.01, 0.05, and 0.20. Selection used the interquartile range of changes, lag-1 autocorrelation, run-length entropy, and spectral entropy. Conditional on the selected dynamics, the measurement search crossed residual standard deviations of 0.10, 0.20, and 0.30 with κ values of 0.75, 1.00, and 1.25. Measurement selection used 16 discrepancy components: median level, level IQR,

change IQR, lag-1 autocorrelation, run-length entropy, spectral entropy, effective rank, and total variance, each for scores and log-ratios. Total variance was the sum of the coordinate variances. The additional entropy and robust variance statistics belonged to the subsequent 22-feature evaluation. Each candidate was evaluated with three calibration simulations. All four folds selected a period of 12 observations, damping ratio 0.20, residual standard deviation 0.30, and $\kappa = 0.75$.

The static logistic-normal generator used the same calibrated coordinate center, covariance geometry, between-video baseline and scale distributions, observation positions, and segment boundaries. It drew independent log-ratio observations within each video. Each selected generator produced 100 datasets per fold with 25 simulated series in each dataset.

Evaluation. The MAFW reference used 1,000 comparisons per fold. Each comparison drew two disjoint, category-stratified groups of 25 videos from the 75 calibration videos. Every feature was standardized by the median and scale of its finite values in the calibration videos. The scale was the IQR, with successive fallbacks to 1.4826 times the median absolute deviation, the standard deviation, and one; a data-derived scale had to exceed 10^{-8}. Before standardization, any undefined video feature was replaced by its calibration median and therefore became zero after standardization. This replacement used no evaluation videos. One MAFW video had no segment of eight observations, leaving its two spectral-entropy features undefined; these were the only two undefined values among the 2,200 empirical video features. The distance between two series was the root-mean-square Euclidean distance across all 22 standardized features, including features replaced in this way. The fold-specific similarity threshold was the 95th percentile of the pooled directional nearest-neighbor distances between the two MAFW groups.

For each generated dataset, plausibility was the proportion of its 25 simulated series whose nearest MAFW evaluation series fell within the threshold. Coverage was the proportion of the 25 MAFW evaluation series whose nearest simulated series fell within the threshold. Results were averaged across the four folds. The reported ranges are the 2.5th and 97.5th percentiles across the 100 generated datasets or 1,000 MAFW reference comparisons.

Distributional comparisons. Calibration and metric-specific distributional discrepancy used finite channel-by-video contributions for each univariate statistic, and one contribution per video for each joint statistic. For univariate statistics, these distributions pooled channel values across videos. Undefined contributions were omitted separately from the empirical and simulated distributions. Their absolute median difference and one-dimensional Wasserstein distance were divided by the calibration scale, using the same scale fallback rule, and the larger standardized value was retained. The measurement objective averaged its 16 components,

giving equal weight to scores and log-ratios; an entirely undefined component made the full selection loss infinite. All stored calibration objectives were finite. The reported full distributional discrepancy averaged the 22 evaluation components.

B.2 Study 2 Simulation Designs

For each clip, LOMM sets or draws the starting state and acceleration inputs according to the configuration, advances each latent dimension, applies the activity schedule, and constructs the observed indicators. The primary, shorter-series, and extended factorial designs use continuous indicators. All clips share the same indicator assignments and nominal measurement parameters. The generator draws clip-specific loading and residual-variance perturbations as specified in the random-draw table below. The one-at-a-time sensitivity conditions use fixed equal measurement parameters except for the stated change.

Table B1: The six Study 2 scenario families. Dwell values are fractions of the number of observations per clip.

Code	Family	Switching	Active / background gain	Dwell fraction	Dynamic rule
A	Weak expression, one dominant dimension	No	0.35 / 0.05	1.00 / 1.00	Stable, low input scale
B	Strong expression, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Stable, standard input scale
C	Strong expression, switching dominant dimension	Yes	1.00 / 0.08	0.20 / 0.45	Stable, standard input scale
D	Strong expression, rapid switching	Yes	1.25 / 0.08	0.03 / 0.10	Stable, high input scale
E	Amplifying dynamics, one dominant dimension	No	1.00 / 0.08	1.00 / 1.00	Positive velocity feedback
F	Amplifying dynamics, switching dominant dimension	Yes	1.10 / 0.10	0.12 / 0.30	Positive velocity feedback

In non-switching scenarios, one dimension remains dominant for a whole clip and the dominant dimension rotates across clips. Switching scenarios sample dwell intervals and change the dominant dimension over time. Families A through D use stable dynamics. Families E and F use positive velocity feedback and test recovery under finite-horizon amplification.

The dynamic parameters are listed in Table B2. Each vector gives one value per latent dimension. The primary and shorter designs use $\Delta = 0.4$; the extended design with 300 observations uses $\Delta = 0.25$. The displayed σ_q values are input standard deviations per transition. Holding them fixed while changing Δ changes the input specification per unit of time.

Table B2: Dynamic parameters for each scenario family. Entries in each vector follow the dimension order. Acceleration-input SD is the per-step standard deviation σ_q. Four-dimensional conditions occur only in the extended factorial simulation.

Code	m	η	ζ	Acceleration-input SD
A	2D	-0.30; -0.55	-1.70; -1.00	0.06; 0.15
A	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.06; 0.15; 0.23
A	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.06; 0.15; 0.23; 0.33
B	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
B	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
B	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
C	2D	-0.30; -0.55	-1.70; -1.00	0.12; 0.30
C	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.12; 0.30; 0.45
C	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.12; 0.30; 0.45; 0.65
D	2D	-0.30; -0.55	-1.70; -1.00	0.15; 0.38
D	3D	-0.30; -0.55; -0.85	-1.70; -1.00; -0.65	0.15; 0.38; 0.56
D	4D	-0.30; -0.55; -0.85; -1.10	-1.70; -1.00; -0.65; -0.30	0.15; 0.38; 0.56; 0.81
E	2D	-0.05; -0.10	0.08; 0.10	0.25; 0.25
E	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.25; 0.25; 0.25
E	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.25; 0.25; 0.25; 0.25
F	2D	-0.05; -0.10	0.08; 0.10	0.35; 0.35
F	3D	-0.05; -0.10; -0.15	0.08; 0.10; 0.12	0.35; 0.35; 0.35
F	4D	-0.05; -0.10; -0.15; -0.20	0.08; 0.10; 0.12; 0.14	0.35; 0.35; 0.35; 0.35

The primary $T = 100$ design contains 21,600 datasets: 216 factorial conditions with 100 paired replications per condition. The factorial crosses six scenario families, $m \in \{2, 3\}$ dimensions, $N \in \{50, 100\}$ clips, loadings $\{0.40, 0.60, 0.80\}$, residual standard deviations $\{0.125, 0.25, 0.50\}$, and four indicators per dimension. The shorter-series design generates 21,600 corresponding datasets with $T = 25$, $\Delta = 0.4$, and duration 9.6. Every method receives the same dataset within each condition and replication.

The one-at-a-time sensitivity analyses fix $N = 100$, $T = 100$, loading 0.60, residual standard deviation 0.25, and four indicators per dimension (the reference condition). They cross all six families, two dimensionalities, and 50 paired replications while changing one feature at a time.

Table B3: One-at-a-time sensitivity conditions. Each changes one design component of the reference condition.

Sensitivity condition	Specification
Alternative initialization	Stationary zero-mean starts for stable dimensions; bounded independent starts for amplifying dimensions
Gradual activity transitions	A five-observation transition replaces each abrupt switch
Cross-loadings	Normal cross-loadings with pre-truncation SD 0.05, truncated to [-0.10, 0.10]
Residual equicorrelation	Correlation 0.10 between every residual pair, including pairs from different dimensions
Correlated inputs	Acceleration-input correlation 0.15 between dimensions
Bounded scores	Direct softmax of the p indicators with tau = 1, analyzed as raw scores or CLR components

Table B4: Random draws and perturbations.

Component	Scope	Specification
Acceleration input	Studies 1 and 2	$\mathbf{q}_{i,n} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D}_q\mathbf{R}_q\mathbf{D}_q)$. The primary designs use $\mathbf{R}_q = \mathbf{I}_m$; the correlated-input sensitivity sets every off-diagonal correlation to 0.15. Draws are independent across clips and transitions. Study 1 scales σ_{qf} to give unit stationary position variance; Table B2 gives the Study 2 values.
Initial state	Study 1	At the start of each uninterrupted segment, each dimension is drawn independently as $\mathbf{s}_{if,1} \sim \mathcal{N}_2(\mathbf{0}, \mathbf{P}_f)$, where $\mathbf{P}_f$ is the stationary covariance for the selected transition and input scale.
Activity	Study 1	Activity modulation is disabled: $a_{if,n} = 1$ for every dimension and generated observation.
Primary loadings	Study 2 matched and extended designs	For item j assigned to dimension f, $\Lambda^{(z)}_{i,jf} = \lambda + U_{ij}$ with independent $U_{ij} \sim \text{Unif}(-0.15, 0.15)$. All cross-loadings are zero.
Residual variances	Study 2 matched and extended designs	For indicator j, $\Psi_{i,jj} = s_\varepsilon^2 + V_{ij}$ with independent $V_{ij} \sim \text{Unif}(-0.01, 0.01)$; off-diagonal entries are zero.
Switching dwell	Study 2 switching families	Set $L_{\min} = \max\{1, \lfloor Td_{\min} \rfloor\}$ and $L_{\max} = \max\{L_{\min}, \lceil Td_{\max} \rceil\}$. Each dwell is sampled uniformly from the integers $L_{\min}, \ldots, L_{\max}$; the final dwell is truncated at observation T. The first dominant dimension rotates across clips, and each later dominant dimension is sampled uniformly from the other $m - 1$ dimensions.
Cross-loadings	Study 2 sensitivity	Each non-primary loading is drawn independently from $\mathcal{N}(0, 0.05^2)$ and truncated componentwise to $[-0.10, 0.10]$. Primary loadings remain fixed at 0.60.
Factor baseline $\boldsymbol{\beta}_i$	Study 2 between-clip analysis	$\boldsymbol{\beta}_i \sim \mathcal{N}_m(\mathbf{0}, 1.8487025519^2\mathbf{I}_m)$ and $\mathbf{b}_i^{(z)} = \boldsymbol{\Lambda}^{(z)}\boldsymbol{\beta}_i$.
Clip scale $s_i^{(z)}$	Study 2 between-clip analysis	$s_i^{(z)} \sim \text{Unif}(0.5, 1.5)$, independently across clips. It multiplies the within-clip signal and measurement residual before the baseline is added.

The between-clip analysis uses 80 paired datasets from stable families A through D, $m \in \{2, 3\}$, and 10 replications. Each dataset has $N = 100$, $T = 100$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and independent stationary starts. Its baseline varies within the loading space according to $\boldsymbol{\beta}_i \sim \mathcal{N}_m(\mathbf{0}, 1.8487025519^2\mathbf{I}_m)$, and its clip scale follows $s_i^{(z)} \sim \text{Unif}(0.5, 1.5)$.

The extended factorial simulation evaluates DynEGA on 518,400 datasets. It crosses $N \in \{50, 100\}$, $T \in \{100, 300\}$, $m \in \{2, 3, 4\}$, $k \in \{4, 6\}$, three loading levels, three residual-error levels, six scenarios, and 400 replications. The $T = 100$ conditions use $\Delta = 0.4$ and the $T = 300$ conditions use $\Delta = 0.25$ while holding per-step input variance fixed. Their difference therefore combines the number of observations, the observation interval, the physical duration, and the discrete input specification.

The public lomm_observe() function implements the observation step and records the configuration, seeds, and hashes needed for reproduction. Study 2 uses the benchmark generator documented in the analysis code, which returns continuous indicators for the primary designs and applies the direct-softmax and CLR transformation for the bounded-score sensitivity condition.

B.3 Method Implementations

DynEGA estimates first derivatives from time-delay embeddings, constructs an EBICglasso partial-correlation network, and groups its positive edges (Boker et al., 2010; H. Golino et al., 2020; H. Golino et al., 2022; H. F. Golino & Epskamp, 2017). Video identity prevents embedding windows from crossing clip boundaries. An explicit search selects the embedding dimension by the Total Entropy Fit Index, with shorter embeddings breaking ties. Candidate dimensions are $\{5, 10, 15, 25, 50\}$ for $T = 100$ and $\{5, 6, 10, 12\}$ for $T = 25$. EGAnet 2.3.0 uses GLLA embedding parameters tau = 1 and delta = 1, first derivatives, automatic correlations, pairwise missing-data handling, and the glasso model. The derivative scale does not affect the correlation matrix because conversion from observation units to physical time multiplies every derivative by the same positive constant.

Static EGA applies EBICglasso to the contemporaneous correlation matrix of the stacked indicator levels and then uses the same network-processing and community-detection rules as DynEGA. It ignores the observation order. This makes it a benchmark that shows how much of the grouping is recoverable without temporal information, and a computational benchmark for the temporal procedures.

GraphicalVAR uses mlGraphicalVAR with clip identity as the grouping variable, within-clip centering, variable standardization, and gamma 0.5 (Epskamp et al., 2018; Epskamp, 2024). Lagged pairs never cross clip boundaries. The grouping uses the returned fixedPCC contemporaneous residual partial-correlation matrix. The diagonal is removed and the matrix is treated as an undirected weighted graph.

GIMME fits each standardized clip separately with unified structural equation modeling and vector autoregression (Gates & Molenaar, 2012; Lane et al., 2019). Standardized residual covariance matrices from successful fits are aligned by variable name, averaged across clips, and symmetrized. The resulting graph is a residual-covariance graph. Unequal residual variances can therefore influence its weights. This conversion was built for the present comparison and differs from typical idiographic use.

For every method, negative graph weights are set to zero. If no positive edge remains, absolute values of the original off-diagonal weights are used; only an all-zero matrix produces singleton communities. The integrated outputs do not record the frequency of this fallback. Louvain at resolution 1.0 is primary (Blondel et al., 2008). Leiden and Walktrap are applied to the same fitted network as sensitivity analyses (Pons & Latapy, 2005; Traag et al., 2019). A deterministic offset from the dataset seed initializes each method fit.

GraphicalVAR and GIMME were omitted from the 518,400-dataset extended factorial simulation because the full design was computationally impractical. Both were included in the primary and shorter-series designs. GraphicalVAR was also included in the one-at-a-time sensitivity and between-clip analyses. GIMME was

included in the between-clip analysis and excluded from the one-at-a-time sensitivity analyses.

B.4 Outcomes, Uncertainty, and Computing

A method can fail to recover the generating dimensions because it returns no usable network or because its network yields the wrong number of groups. The primary recovery rate counts both outcomes as failures and uses all attempted datasets as its denominator. We also report the proportion of attempts that return a valid network and the proportion of valid outputs with the correct number of dimensions. These supporting rates separate computational failure from incorrect grouping.

Adjusted Rand index and variation of information compare the full estimated and prespecified partitions among valid outputs. Higher adjusted Rand index and lower variation of information indicate closer agreement. Assignment accuracy matches estimated groups to dimensions to maximize correct indicator assignments. Every indicator remains in the denominator, and indicators in unmatched groups count as incorrect.

Trajectory diagnostics assess whether the recovered groups summarize the time course of the generating processes. They are constructed from the levels of the analyzed indicators after grouping, including for DynEGA. Scores retain the original observation order within each clip. The indicators are standardized over the pooled clips, and each recovered group is summarized by its first principal component. A single-indicator group uses its standardized indicator, and a row mean is used if a component cannot be computed. Each resulting summary is then standardized over the pooled clips. Let $\mathbf{y}_{ig}$ denote the summary of recovered group g in clip i, and let $\tilde{\mathbf{x}}_{if}$ denote the activity-scaled trajectory of data-generating dimension f. The best-match absolute trajectory correlation is

$$R_{\text{best}} = \frac{1}{Nm} \sum_{i=1}^{N} \sum_{f=1}^{m} \max_{g} \left| \text{cor} \left(\tilde{\mathbf{x}}_{if}, \mathbf{y}_{ig} \right) \right| .$$

Correlations use pairwise complete observations. The maximizing recovered group is selected separately for each data-generating dimension, so the same recovered-group summary may be selected for multiple dimensions.

Normalized root mean squared error (NRMSE) measures the difference between the standardized trajectories. It uses a separate one-to-one assignment within each clip. For data-generating dimension f and recovered group g, the sign-invariant cost is

$$d_{ifg} = \min_{\delta \in \{-1,1\}} \left[\frac{1}{K_{ifg}} \sum_{n \in \mathcal{O}_{ifg}} \left\{ \frac{\tilde{x}_{if,n} - \bar{x}_{if}}{s_{x,if}} - \delta \frac{y_{ig,n} - \bar{y}_{ig}}{s_{y,ig}} \right\}^2 \right]^{1/2},$$

where $\mathcal{O}_{ifg}$ contains the K_{ifg} observations for which both standardized series are finite. The means and sample standard deviations are computed separately for each series within each clip. Let $\mathcal{O}_{x,if}$ contain the $K_{x,if}$ finite observations of the activity-scaled latent trajectory. Its sample variance is

$$s^2_{x,if} = \frac{1}{K_{x,if} - 1} \sum_{n \in \mathcal{O}_{x,if}} \left(\tilde{x}_{if,n} - \bar{x}_{if} \right)^2,$$

and $s^2_{y,ig}$ is defined analogously for the recovered-group summary. Thus, NRMSE is the root mean squared difference between two within-clip standardized series after choosing the sign that gives the smaller value.

For each clip, a linear assignment algorithm selects the one-to-one matching $\widehat{\pi}_i$ that minimizes the total cost. A comparison with fewer than three finite pairs after standardization, including comparisons involving a constant series, receives a cost of 2. The cost matrix is padded with a value of 2 when a data-generating dimension has no recovered-group match. The dataset-level diagnostic is

$$\text{NRMSE} = \frac{1}{Nm} \sum_{i=1}^{N} \sum_{f=1}^{m} d_{if,\widehat{\pi}_i(f)}.$$

Correlation allows several latent dimensions to share the same recovered summary; NRMSE assigns a different summary to each matched dimension. All-attempt failure-adjusted versions assign correlation 0 and NRMSE 2 when no valid network output exists or the corresponding dataset diagnostic is undefined.

Uncertainty uses 2,000 bootstrap resamples of whole datasets within each reported analysis group. Monte Carlo standard errors use delete-one-dataset jackknifing. The three community-detection algorithms reuse the same fitted structure and therefore assess the grouping step rather than estimator variability.

Analyses used R 4.5.1 on Rocky Linux 8.8 with dual Intel Xeon Silver 4310 processors, 48 logical threads, and 125 GiB RAM. Key versions were EGAnet 2.3.0, graphicalVAR 0.3.4, gimme 10.0, igraph 2.2.1, qgraph 1.9.8, psych 2.5.6, lavaan 0.6-20, dplyr 1.1.4, and ggplot2 4.0.1. GraphicalVAR and GIMME used a 900-second limit per fit in the primary and shorter designs.

Runtime is elapsed wall-clock time for attempts returning a valid network output. It includes structure estimation, network extraction, and community detection. One fitted structure is reused for Louvain, Leiden, and Walktrap, so feasibility counts unique fits and accuracy counts fit-and-algorithm results. Runtime values

are specific to this implementation and machine.

Appendix C: Supplementary Results

C.1 Metric-Level Results for Study 1

Figure C1 compares LOMM with the MAFW-to-MAFW reference for each of the 22 features. Fourteen LOMM discrepancies fell within the reference range. The full 22-feature distributional discrepancy was 0.2543 [0.2356, 0.2738] for LOMM and 0.1811 [0.1521, 0.2193] for the MAFW reference. The corresponding median for the static logistic-normal generator was 0.6942.

LOMM closely reproduced most measures of level, dispersion, short-term change, persistence, effective dimensionality, and robust variation. Its largest departures occurred for entropy. Sample entropy was higher and spectral entropy was lower than in MAFW for both scores and log-ratio coordinates. Log-ratio permutation entropy was lower, effective rank was higher, and robust level variance was lower. The score median-level discrepancy was just outside the reference range, although the signed median difference was 0.004.

Diagnostic subsets gave the same interpretation. A time-domain subset containing level, spread, change, lag-1 autocorrelation, and run-length entropy had a LOMM discrepancy of 0.1463, within the MAFW reference interval [0.0967, 0.1489]. A local-temporal subset containing change, lag-1 autocorrelation, and run-length entropy also fell within its reference interval. The two robust variance measures together fell within the MAFW reference range. The entropy measures accounted for most of the remaining full-panel discrepancy.

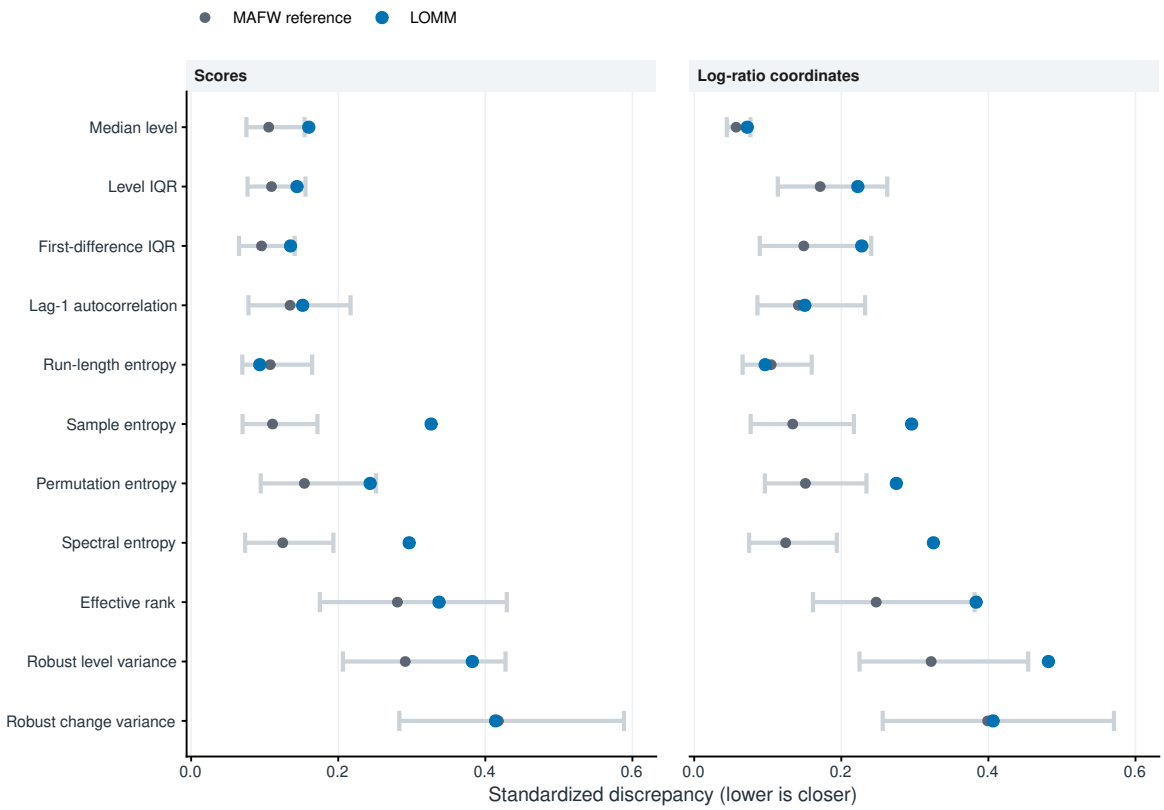


Figure C1: Metric-level discrepancies between LOMM and MAFW classifier-score series. Blue points are medians across 100 LOMM simulations. Gray points and bars show the median and central 95% reference range across 1,000 disjoint MAFW-to-MAFW comparisons. Results are averaged across four folds and shown for scores and log-ratio representations. Discrepancy is the larger of the standardized absolute median difference and standardized Wasserstein distance. Smaller values indicate closer agreement.

C.2 Grouping and Trajectory Diagnostics

This appendix reports two types of post-estimation diagnostic. Grouping diagnostics evaluate recovered indicator groupings among valid network outputs. Trajectory diagnostics assess alignment between recovered-group summaries and activity-scaled latent trajectories; the trajectories are not direct latent-state estimates produced by the methods.

Table C1: Louvain grouping diagnostics for the matched-method simulation with $T = 100$. All diagnostics use valid network outputs. Signed dimension error is the estimated number of dimensions minus the generating number; absolute error is its magnitude. Higher adjusted Rand index and assignment accuracy, and lower variation of information, indicate closer agreement. MCSE denotes Monte Carlo standard error. The 95% intervals resample whole datasets within each reported analysis group.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid outputs	Correct dimensions
DynEGA	Signed dimension error	0.166	0.0057	[0.154, 0.177]	21600	21600	20282
DynEGA	Absolute dimension error	0.168	0.0057	[0.157, 0.180]	21600	21600	20282
DynEGA	Adjusted Rand index	0.935	0.0015	[0.932, 0.938]	21600	21600	20282
DynEGA	Variation of information	0.095	0.0023	[0.091, 0.100]	21600	21600	20282
DynEGA	Assignment accuracy	0.965	0.0009	[0.963, 0.966]	21600	21600	20282
GraphicalVAR	Signed dimension error	0.979	0.0171	[0.947, 1.011]	20329	20329	16784
GraphicalVAR	Absolute dimension error	0.979	0.0171	[0.945, 1.012]	20329	20329	16784
GraphicalVAR	Adjusted Rand index	0.867	0.0022	[0.863, 0.872]	20329	20329	16784
GraphicalVAR	Variation of information	0.183	0.0031	[0.177, 0.189]	20329	20329	16784
GraphicalVAR	Assignment accuracy	0.902	0.0016	[0.899, 0.906]	20329	20329	16784
GIMME	Signed dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Absolute dimension error	0.319	0.0110	[0.298, 0.341]	4846	4846	3795
GIMME	Adjusted Rand index	0.733	0.0057	[0.722, 0.744]	4846	4846	3795
GIMME	Variation of information	0.337	0.0073	[0.323, 0.351]	4846	4846	3795
GIMME	Assignment accuracy	0.877	0.0027	[0.872, 0.882]	4846	4846	3795
Static EGA	Signed dimension error	0.799	0.0159	[0.770, 0.829]	21526	21526	19093
Static EGA	Absolute dimension error	0.799	0.0159	[0.767, 0.829]	21526	21526	19093
Static EGA	Adjusted Rand index	0.894	0.0021	[0.889, 0.898]	21526	21526	19093
Static EGA	Variation of information	0.147	0.0029	[0.142, 0.153]	21526	21526	19093
Static EGA	Assignment accuracy	0.922	0.0015	[0.919, 0.925]	21526	21526	19093

Table C2: Louvain grouping diagnostics for the matched-method simulation with $T = 25$. All diagnostics use valid network outputs. Signed dimension error is the estimated number of dimensions minus the generating number; absolute error is its magnitude. Higher adjusted Rand index and assignment accuracy, and lower variation of information, indicate closer agreement. MCSE denotes Monte Carlo standard error. The 95% intervals resample whole datasets within each reported analysis group.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid outputs	Correct dimensions
DynEGA	Signed dimension error	0.492	0.0102	[0.471, 0.512]	21600	21600	18636
DynEGA	Absolute dimension error	0.494	0.0102	[0.474, 0.515]	21600	21600	18636
DynEGA	Adjusted Rand index	0.902	0.0018	[0.899, 0.906]	21600	21600	18636
DynEGA	Variation of information	0.137	0.0025	[0.132, 0.143]	21600	21600	18636
DynEGA	Assignment accuracy	0.935	0.0012	[0.933, 0.938]	21600	21600	18636
GraphicalVAR	Signed dimension error	1.048	0.0154	[1.017, 1.076]	21579	21579	16641
GraphicalVAR	Absolute dimension error	1.048	0.0154	[1.016, 1.080]	21579	21579	16641
GraphicalVAR	Adjusted Rand index	0.859	0.0020	[0.855, 0.863]	21579	21579	16641
GraphicalVAR	Variation of information	0.200	0.0028	[0.195, 0.206]	21579	21579	16641
GraphicalVAR	Assignment accuracy	0.893	0.0015	[0.890, 0.896]	21579	21579	16641
Static EGA	Signed dimension error	0.576	0.0136	[0.551, 0.603]	21541	21541	19721
Static EGA	Absolute dimension error	0.576	0.0136	[0.550, 0.603]	21541	21541	19721
Static EGA	Adjusted Rand index	0.922	0.0018	[0.919, 0.926]	21541	21541	19721
Static EGA	Variation of information	0.108	0.0025	[0.103, 0.113]	21541	21541	19721
Static EGA	Assignment accuracy	0.943	0.0013	[0.940, 0.945]	21541	21541	19721

Table C3: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 100$. Rows marked correct dimensions include valid outputs that recovered the correct number of dimensions. Absolute correlation denotes best-match absolute trajectory correlation; NRMSE denotes normalized root mean squared error. Correlation may reuse a recovered-group summary for multiple latent dimensions; NRMSE uses a separate one-to-one match. All-attempt rows assign correlation 0 and NRMSE 2 when no valid network output was returned or the corresponding diagnostic was undefined. Higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid	Correct dimension
DynEGA	Absolute correlation (all attempts)	0.728	0.0017	[0.725, 0.731]	21600	21600	20282
DynEGA	NRMSE (all attempts)	0.642	0.0031	[0.636, 0.647]	21600	21600	20282
GraphicalVAR	Absolute correlation (all attempts)	0.668	0.0020	[0.664, 0.671]	21600	20329	16784
GraphicalVAR	NRMSE (all attempts)	0.753	0.0036	[0.746, 0.760]	21600	20329	16784
GIMME	Absolute correlation (all attempts)	0.132	0.0019	[0.129, 0.136]	21600	4846	3795
GIMME	NRMSE (all attempts)	1.749	0.0034	[1.742, 1.756]	21600	4846	3795
Static EGA	Absolute correlation (all attempts)	0.727	0.0017	[0.723, 0.730]	21600	21526	19093
Static EGA	NRMSE (all attempts)	0.641	0.0031	[0.635, 0.647]	21600	21526	19093
DynEGA	Absolute correlation (correct dimensions)	0.760	0.0015	[0.757, 0.763]	20282	21600	20282
DynEGA	NRMSE (correct dimensions)	0.602	0.0031	[0.596, 0.607]	20282	21600	20282
GraphicalVAR	Absolute correlation (correct dimensions)	0.791	0.0014	[0.789, 0.794]	16784	20329	16784
GraphicalVAR	NRMSE (correct dimensions)	0.568	0.0031	[0.562, 0.574]	16784	20329	16784
GIMME	Absolute correlation (correct dimensions)	0.606	0.0034	[0.600, 0.613]	3795	4846	3795
GIMME	NRMSE (correct dimensions)	0.898	0.0050	[0.888, 0.907]	3795	4846	3795
Static EGA	Absolute correlation (correct dimensions)	0.787	0.0014	[0.784, 0.789]	19093	21526	19093
Static EGA	NRMSE (correct dimensions)	0.563	0.0031	[0.557, 0.569]	19093	21526	19093

Table C4: Louvain trajectory-summary diagnostics for the matched-method simulation with $T = 25$. Rows marked correct dimensions include valid outputs that recovered the correct number of dimensions. Absolute correlation denotes best-match absolute trajectory correlation; NRMSE denotes normalized root mean squared error. Correlation may reuse a recovered-group summary for multiple latent dimensions; NRMSE uses a separate one-to-one match. All-attempt rows assign correlation 0 and NRMSE 2 when no valid network output was returned or the corresponding diagnostic was undefined. Higher correlation and lower NRMSE indicate closer alignment with the activity-scaled latent trajectories.

Method	Diagnostic	Estimate	MCSE	95% interval	Datasets	Valid	Correct dimension
DynEGA	Absolute correlation (all attempts)	0.800	0.0012	[0.797, 0.802]	21600	21600	18636
DynEGA	NRMSE (all attempts)	0.593	0.0023	[0.589, 0.598]	21600	21600	18636
GraphicalVAR	Absolute correlation (all attempts)	0.799	0.0012	[0.796, 0.801]	21600	21579	16641
GraphicalVAR	NRMSE (all attempts)	0.587	0.0023	[0.582, 0.591]	21600	21579	16641
GIMME	Absolute correlation (all attempts)	0.000	0.0000	[0.000, 0.000]	21600	0	0
GIMME	NRMSE (all attempts)	2.000	0.0000	[2.000, 2.000]	21600	0	0
Static EGA	Absolute correlation (all attempts)	0.801	0.0012	[0.799, 0.803]	21600	21541	19721
Static EGA	NRMSE (all attempts)	0.595	0.0024	[0.591, 0.600]	21600	21541	19721
DynEGA	Absolute correlation (correct dimensions)	0.848	0.0009	[0.846, 0.850]	18636	21600	18636
DynEGA	NRMSE (correct dimensions)	0.521	0.0023	[0.517, 0.526]	18636	21600	18636
GraphicalVAR	Absolute correlation (correct dimensions)	0.871	0.0008	[0.870, 0.873]	16641	21579	16641
GraphicalVAR	NRMSE (correct dimensions)	0.466	0.0022	[0.461, 0.470]	16641	21579	16641
Static EGA	Absolute correlation (correct dimensions)	0.836	0.0009	[0.834, 0.838]	19721	21541	19721
Static EGA	NRMSE (correct dimensions)	0.548	0.0023	[0.544, 0.553]	19721	21541	19721

C.3 Sensitivity Analyses and Realized Amplification

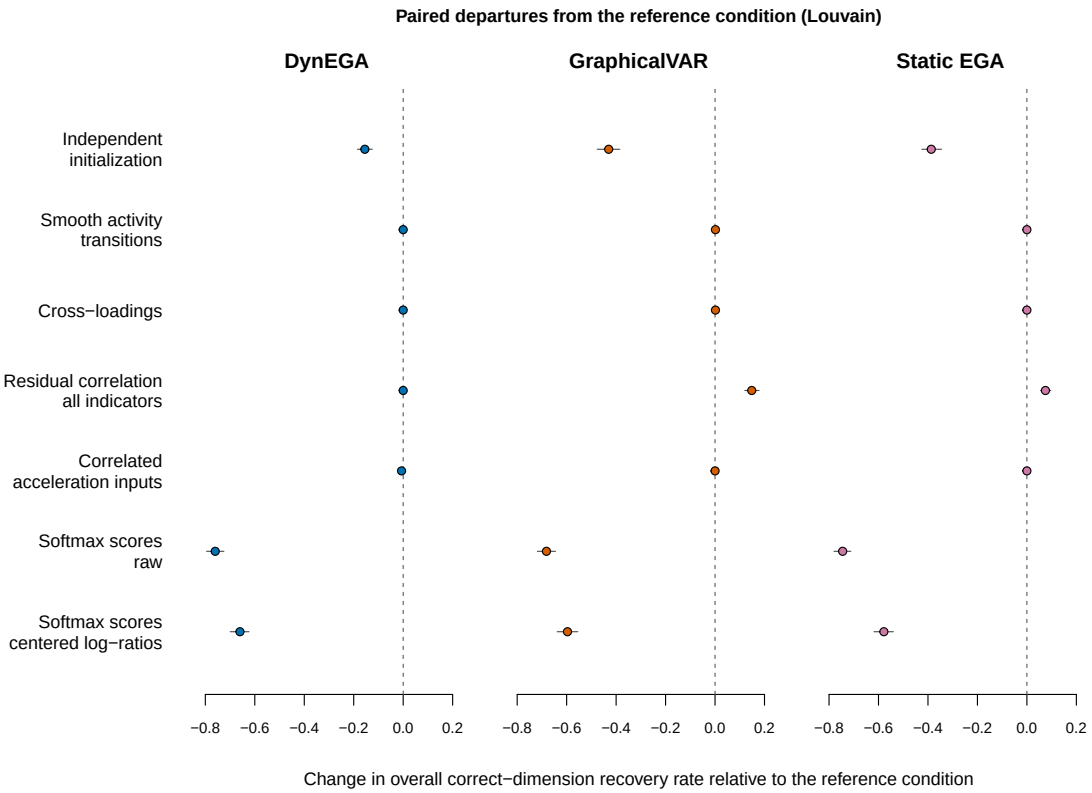


Figure C2: Paired change in overall correct-dimension recovery under each one-at-a-time sensitivity condition. Negative values indicate lower recovery than in the reference condition. Error bars are 95% bootstrap intervals from resampling whole datasets.

Table B2 gives the dynamic parameters used in these scenarios. Table C5 summarizes the amplitudes generated in the amplifying families E and F. Table C6 describes deterministic growth or decay for each distinct parameter pair. Together, these diagnostics distinguish the specified dynamic regime from the amplitudes observed in finite simulated records.

Table C5: Realized amplitude diagnostics for the amplifying families E and F (Table B1) with $N = 50$, four indicators per dimension, loading 0.60, residual standard deviation 0.25, and 25 replications. The final-tenth energy fraction is the proportion of squared activity-scaled latent signal occurring in the last 10% of observations, averaged across clips within a panel and then summarized across replications.

Code	Dimensions	T	H	Median max $\lvert\tilde{x}\rvert$	95th percentile max $\lvert\tilde{x}\rvert$	Median observed IQR	Final-tenth energy fraction
E	2	100	39.60	55.9	61.5	3.47	0.316
E	2	300	74.75	329.4	376.9	7.46	0.338
E	3	100	39.60	62.6	73.0	1.90	0.368

Code	Dimensions	T	H	Median max $\lvert\tilde{x}\rvert$	95th percentile max $\lvert\tilde{x}\rvert$	Median observed IQR	Final-tenth energy fraction
E	3	300	74.75	430.3	478.6	4.58	0.428
E	4	100	39.60	85.6	102.7	1.57	0.404
E	4	300	74.75	994.9	1092.7	4.02	0.498
F	2	100	39.60	69.5	81.9	4.71	0.319
F	2	300	74.75	388.1	491.3	9.31	0.324
F	3	100	39.60	77.7	87.5	2.33	0.367
F	3	300	74.75	518.5	608.5	5.74	0.448
F	4	100	39.60	105.2	122.2	1.92	0.402
F	4	300	74.75	1201.6	1350.2	4.97	0.517

Table C6: Deterministic stability diagnostics for the unique dynamic-parameter pairs used across scenario factors. Table B2 maps these pairs to scenarios and dimensions. The spectral radius $\rho(\Phi)$ is the largest absolute eigenvalue of the transition matrix; values below one indicate decay. The modal exponential-rate factors $g(H) = \exp\{H \max \mathrm{Re}(r)\}$ describe modal rates over the observed horizon rather than total finite-horizon state amplification. $H[100] = 39.6$ and $H[300] = 74.75$ under the stated record designs.

Scenario codes	eta	zeta	Discriminant	Max Re(root)	rho(Phi[.4])	rho(Phi[.25])	g(H[100])	g(H[300])	Regime
A, B, C, D	-1.10	-0.30	-4.3100	-0.150	0.9418	0.9632	0.0026	0.0000	Stable oscillatory
A, B, C, D	-0.85	-0.65	-2.9775	-0.325	0.8781	0.9220	0.0000	0.0000	Stable oscillatory
A, B, C, D	-0.55	-1.00	-1.2000	-0.500	0.8187	0.8825	0.0000	0.0000	Stable oscillatory
A, B, C, D	-0.30	-1.70	1.6900	-0.200	0.9231	0.9512	0.0004	0.0000	Stable nonoscillatory
E, F	-0.20	0.14	-0.7804	0.070	1.0284	1.0177	15.9906	187.2604	Finite-horizon amplifying oscillatory
E, F	-0.15	0.12	-0.5856	0.060	1.0243	1.0151	10.7618	88.6770	Finite-horizon amplifying oscillatory
E, F	-0.10	0.10	-0.3900	0.050	1.0202	1.0126	7.2427	41.9929	Finite-horizon amplifying oscillatory
E, F	-0.05	0.08	-0.1936	0.040	1.0161	1.0101	4.8744	19.8857	Finite-horizon amplifying oscillatory

C.4 Extended Factorial Simulation

The extended factorial simulation evaluated DynEGA on 518,400 datasets. It recovered the correct dimension in 93.7% of those datasets and assigned 95.2% of indicators correctly. Stronger loadings and smaller residual error improved recovery. The design, including the $T = 300$ change in observation interval, is given in Appendix B. Runtime accounting for this design is in Section C.5.

C.5 Runtime and Feasibility Accounting

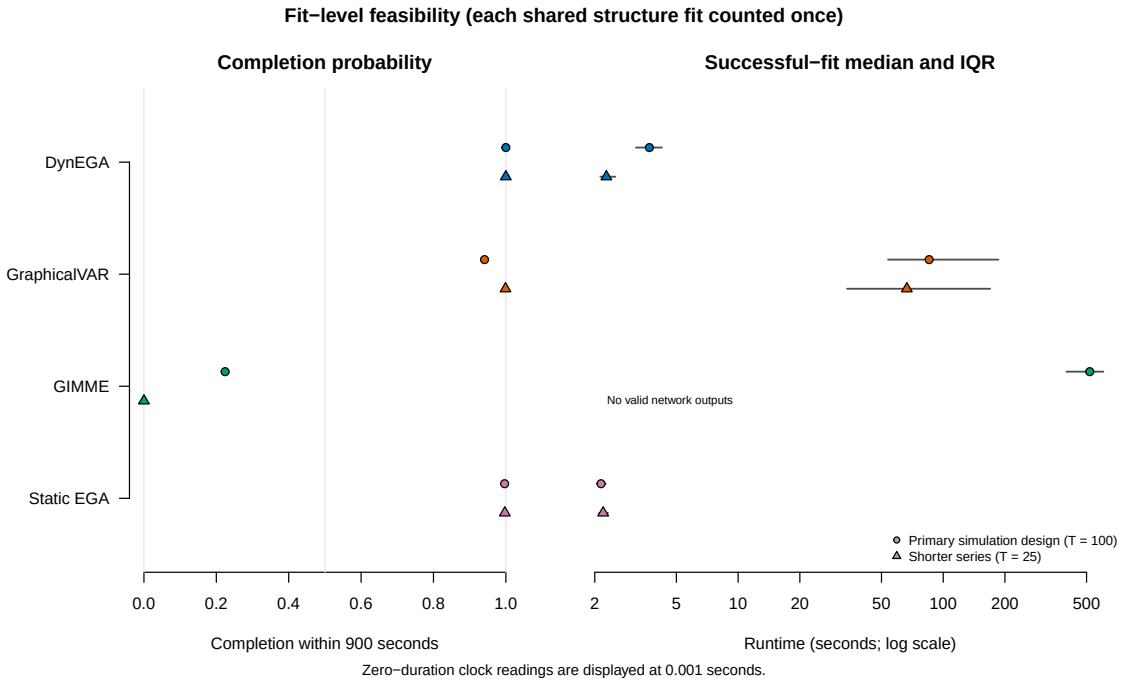


Figure C3: Runtime among valid network outputs in the primary $T = 100$ and shorter $T = 25$ designs. Points show medians and intervals show interquartile ranges on a log10 scale. Counts use unique fitted structures.

Table C7: Runtime and feasibility per method fit in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Valid-output intervals resample whole simulated datasets. Runtime summaries condition on valid network outputs and include structure estimation, network extraction, and community detection. Each fitted structure is reused for Louvain, Leiden, and Walktrap.

Analysis	T	Method	Attempts	Valid outputs	Timeouts	Errors	Valid-output rate	Runtime median [IQR], s
Matched methods	$T = 100$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	3.69 [3.18, 4.25]
Matched methods	$T = 100$	GIMME	21600	4846	14477	2277	0.224 [0.219, 0.230]	518.16 [398.98, 603.71]
Matched methods	$T = 100$	GraphicalVAR	21600	20329	1271	0	0.941 [0.938, 0.944]	85.41 [53.82, 185.91]
Matched methods	$T = 100$	Static EGA	21600	21526	0	74	0.997 [0.996, 0.997]	2.15 [2.05, 2.26]
Matched methods	$T = 25$	DynEGA	21600	21600	0	0	1.000 [1.000, 1.000]	2.28 [2.13, 2.52]
Matched methods	$T = 25$	GIMME	21600	0	0	21600	0.000 [0.000, 0.000]	Not available
Matched methods	$T = 25$	GraphicalVAR	21600	21579	21	0	0.999 [0.999, 0.999]	66.49 [33.96, 169.39]
Matched methods	$T = 25$	Static EGA	21600	21541	0	59	0.997 [0.997, 0.998]	2.20 [2.10, 2.32]

C.6 Community-Detection Sensitivity

Louvain, Leiden, and Walktrap were applied to each retained fitted graph. Their results were closely aligned in the matched-method simulation designs (Figure C4) and led to the same interpretation of the one-at-a-time sensitivity conditions (Figure C5). These comparisons reuse fitted structures and therefore assess the community-detection step rather than estimator variability.

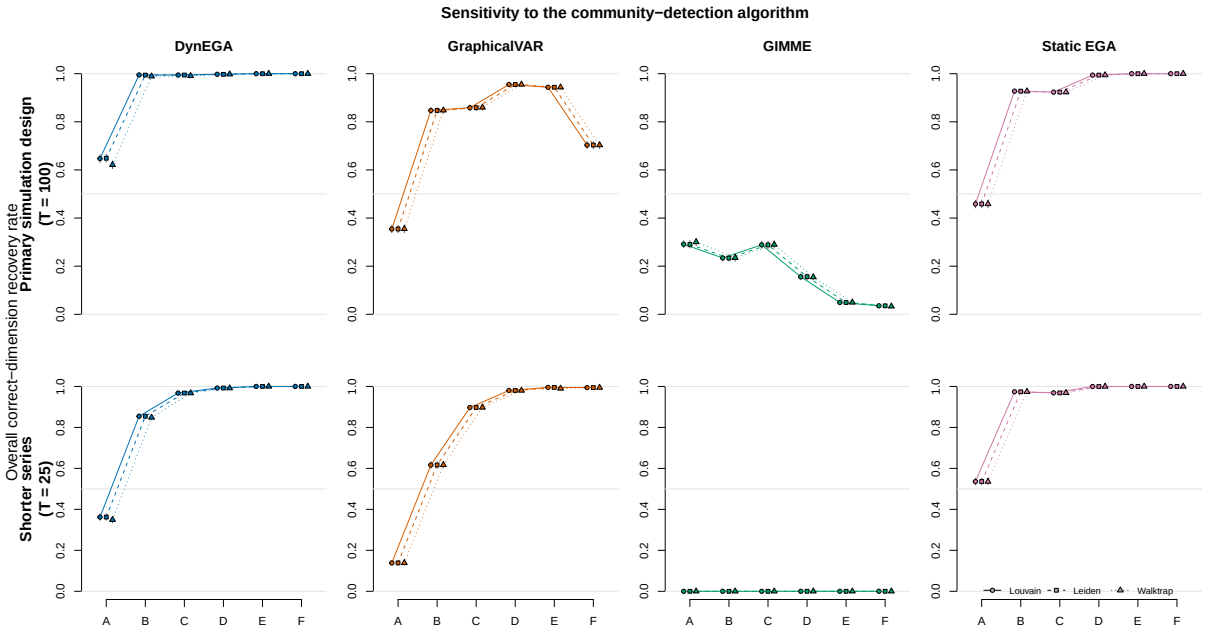


Figure C4: Overall correct-dimension recovery rates for Louvain, Leiden, and Walktrap in the $T = 100$ primary and $T = 25$ shorter-series matched-method designs. Each fitted network is reused across all three algorithms, so differences isolate the community-detection step.

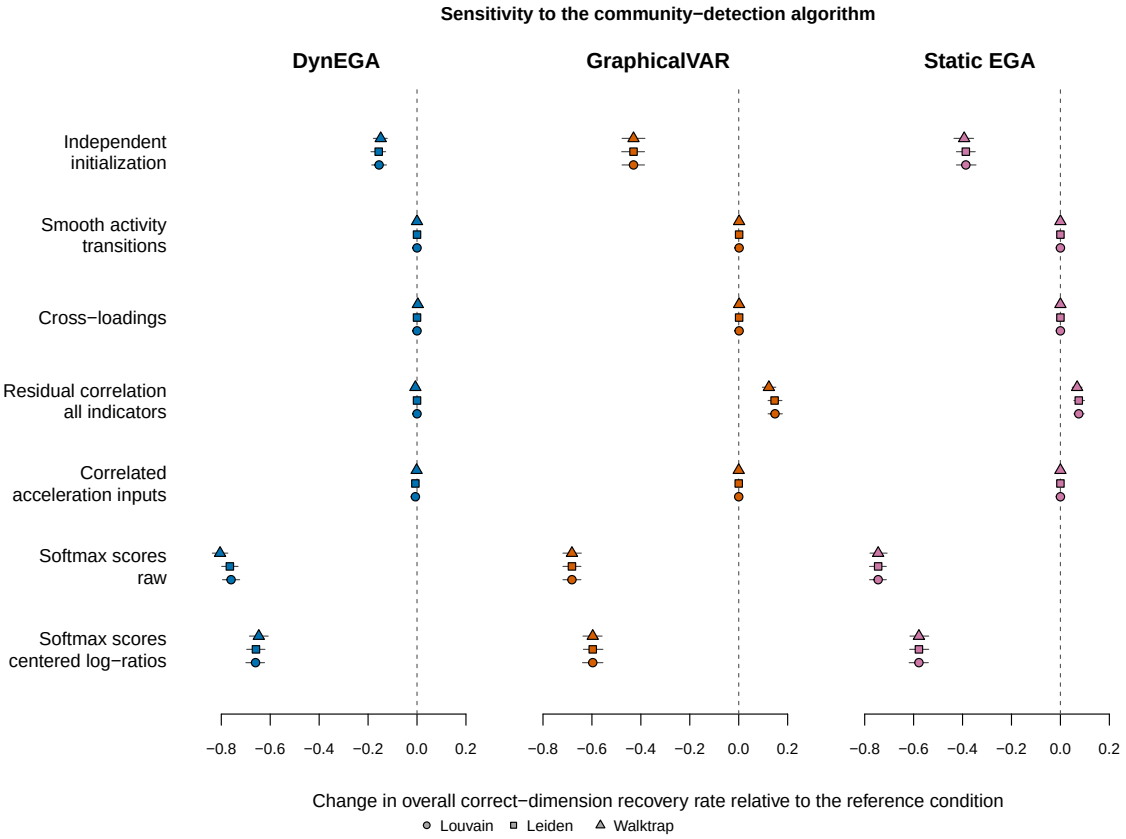


Figure C5: Sensitivity of the paired one-at-a-time effects to Louvain, Leiden, and Walktrap. Values are changes in overall correct-dimension recovery rate relative to the reference condition.

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